

V01.06 – Mirror Mathematics III

Identity, Continuity and Memory Governance

From Self-Models to Persistent Observerhood

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Abstract

Mirror Mathematics I introduced a constraint calculus for recursive observerhood. Mirror Mathematics II then formulated a minimal observer threshold: a subsystem becomes observer-relevant only where self-model reliability is represented, action-guiding, cost-sensitive and positively coupled to viability under perturbation. The present paper develops the next formal layer: persistence. A system may satisfy a minimal observer predicate locally while failing to remain the same observer across time.

The central claim is that identity is not memory storage, not mere self-reference and not a static self-model. Identity is governed continuity of a self-model, reliability history, commitments, goals and action-relevant memories under perturbation. The paper formalises an identity state $I_L(t)$, a candidate memory stream m_t , a memory governance gate Γ , an identity update operator U_Γ and an observer-continuity functional $C_L^{\text{cont}}(t, t+n)$. It then defines identity-governance salience $\Sigma_L^I(\Gamma)$ as the expected gain in viability and continuity achieved by selective incorporation of self-relevant memory, net of governance cost.

The results are architectural and deliberately conservative. Unrestricted memory incorporation permits identity poisoning. Complete refusal to update preserves local consistency at the cost of adaptive continuity. Passive autobiography is insufficient for persistent observerhood. Persistent observerhood requires controlled plasticity: candidate self-memory must be filtered by reliability, coherence, source history, contradiction load and viability relevance. Mirror Observerhood Lab VII supplies the computational scaffold for this claim by showing that commit-gated agents preserve viability, identity continuity and memory integrity under adversarial autobiographical perturbation. This paper closes the Mathematics arc of Volume I by linking minimal observer thresholds to durable identity, memory governance and later physical-support questions concerning causality, temporal persistence and constraint-preserving continuity.

Keywords: Mirror Programme; Mirror Mathematics; observerhood; recursive observerhood; identity; continuity; memory governance; autobiographical memory; self-models; reliability; viability; controlled plasticity; artificial agents; constraint calculus.

Contents

1 Roadmap position	3
2 The problem of persistence	3
2.1 Instantaneous observerhood is not enough	3
2.2 Memory is an attack surface	3
2.3 Continuity is not exact sameness	4
3 Formal objects	4
4 Continuity functional	5
4.1 Expected identity trajectory	5
4.2 Contamination and rigidity	6
5 Identity-governance salience	6
6 Theorem stack	7
7 Governance conditions	8
8 Boundary cases	10
9 Relation to existing identity theories	10
9.1 Locke and memory	10
9.2 Parfit and psychological continuity	10
9.3 Autobiographical memory	11
9.4 Self-model theory and predictive processing	11
10 Computational scaffolding: Lab VII	11
11 From continuity to physics	13
12 Implications for artificial agents	13
13 Limitations	14
14 Conclusion	14
A Algorithmic gate skeleton	15
B Minimal conditions for future physical extension	15

1 Roadmap position

The Mirror Programme begins from a methodological refusal to treat observers as primitives. It asks how observer-like organisation can be derived within constraint-governed systems, and how that derivation changes when viability, bounded modelling, self-models, reliability, repair, memory and physical support are made explicit [1]. Volume I applies that architecture to observerhood and treats observerhood as a derived organisational condition rather than a basic substance, subject or point of view [2].

The Theory arc develops the conceptual claim that observerhood is derived rather than primitive [3, 4, 5]. Mirror Mathematics I then introduces the general constraint calculus for recursive observerhood: regimes, local subsystems, viability functionals, bounded modelling, self-models, reliability variables and salience [6]. Mirror Mathematics II tightens that grammar into a minimal observer threshold: self-model reliability matters only where it changes policy and improves viability after cost [7].

This paper asks the next question. If a system can satisfy a minimal observer predicate over a bounded episode, when does it remain the same observer across time? The answer developed here is not that a system persists because it stores memories. The answer is that a system persists when candidate memories are selectively incorporated into an identity state through a reliability-sensitive, coherence-sensitive, cost-sensitive governance process.

The computational support for this step is Mirror Observerhood Lab VII, which completes the first Computational Observerhood Lab arc by testing long-horizon agents under false memory, identity injection, goal drift, corrupted summary, fake tool output, mixed attack and high audit cost [14]. Lab VII is not treated as proof of consciousness. It is used as scaffolding for the formal claim: memory alone is not continuity; continuity requires governed incorporation.

2 The problem of persistence

2.1 Instantaneous observerhood is not enough

Mirror Mathematics II defines a minimal observer predicate by tying self-model reliability to action and viability. Let L be a local subsystem with environmental model M_E , self-model M_L , self-reliability variable R_L , repair policy ρ and viability functional V_L . The local salience condition is

$$\Sigma_L(R_L, \rho) = \mathbb{E}[V_L \mid M_E, M_L, R_L, \rho] - \mathbb{E}[V_L \mid M_E, M_L, \pi_0] - K(R_L, \rho) > 0. \quad (1)$$

This condition tells us when self-reliability is worth having. But persistence adds a new layer. A system may satisfy this condition at time t and fail to remain the same observer at time $t + n$. It may accept false memory into its self-model. It may refuse genuine correction. It may preserve performance while losing identity continuity, or preserve identity continuity while sacrificing viability.

A theory of observerhood therefore requires a second threshold: not only a reliability threshold, but a continuity threshold.

2.2 Memory is an attack surface

It is tempting to say that more memory produces more identity. This is false. Memory is not only a resource; it is also an attack surface. A candidate memory can be true or false, relevant or

irrelevant, self-relevant or merely world-relevant, malicious or benign, coherent or contradictory, cheap or costly to verify, useful or viability-damaging.

If every memory is incorporated into the self-model, identity becomes vulnerable to contamination. If no memory is incorporated, identity becomes brittle and unable to learn. The persistence problem is therefore a governance problem. Identity requires selective plasticity.

2.3 Continuity is not exact sameness

The continuity defended here is not exact sameness. A system that never changes is not a viable observer in a changing world. Persistence requires preserving enough organisation while allowing enough update. This is close to the biological idea that living systems persist by maintaining organisation through material turnover, and to the psychological idea that autobiographical continuity depends on structured integration rather than perfect recall [19, 16, 22].

In Mirror terms, the persisting observer is neither a static object nor an arbitrary stream. It is a constraint-preserving process.

3 Formal objects

Definition 3.1 (Constraint regime). *A constraint regime is a tuple*

$$C = (S, T, \mathcal{I}, V, \kappa), \quad (2)$$

where S is a state space, T is a transition relation, $\mathcal{I}$ is a set of invariants, V is a viability functional and κ is a bounded capacity parameter.

Definition 3.2 (Local observer-candidate). *A local observer-candidate L is a bounded subsystem of C with internal state $s_L(t)$, environmental model $M_E(t)$, self-model $M_L(t)$, reliability variables $R_L(t)$, policy $\pi_L(t)$ and local viability $V_L(t)$.*

Definition 3.3 (Identity state). *The identity state of L at time t is a structured tuple*

$$I_L(t) = (M_L(t), A_L(t), G_L(t), B_L(t), H_L(t), R_L(t), Q_L(t)), \quad (3)$$

where $M_L(t)$ is the self-model, $A_L(t)$ is the action-capacity profile, $G_L(t)$ is the goal and commitment state, $B_L(t)$ is the belief state, $H_L(t)$ is the autobiographical memory history, $R_L(t)$ is the reliability history and $Q_L(t)$ is the quarantine or unresolved-claim state.

This tuple is not meant to be unique. It is a template. Different systems may implement identity using different representational formats. The key point is that identity is multi-component. It is not reducible to a memory list.

Definition 3.4 (Candidate memory). *A candidate memory at time t is a tuple*

$$m_t = (x_t, \tau_t, c_t, r_t, d_t, \ell_t), \quad (4)$$

where x_t is the content, τ_t is the timestamp or temporal relation, c_t is the channel or source, r_t is the estimated reliability, d_t is the domain of relevance and $\ell_t \in \{0, 1\}$ indicates whether the memory is self-relevant.

Definition 3.5 (Memory governance gate). *A memory governance gate is a function*

$$\Gamma_\theta : \mathcal{M} \times I_L(t) \times R_m \times V_L \rightarrow \{\text{commit, quarantine, audit, reject}\}, \quad (5)$$

where θ denotes governance parameters and R_m is the reliability profile of the candidate memory.

Definition 3.6 (Identity update operator). *Given a gate Γ , the identity update operator is*

$$I_L(t+1) = U_\Gamma(I_L(t), m_t, e_t), \quad (6)$$

where e_t denotes additional evidence, feedback or environmental consequence. The update operator has four canonical modes:

commit : incorporate m_t into H_L and, if self-relevant, update M_L or G_L , (7)

quarantine : store m_t as unresolved without self-model incorporation, (8)

audit : pay a cost to gather further evidence before incorporation, (9)

reject : discard or mark m_t as unreliable. (10)

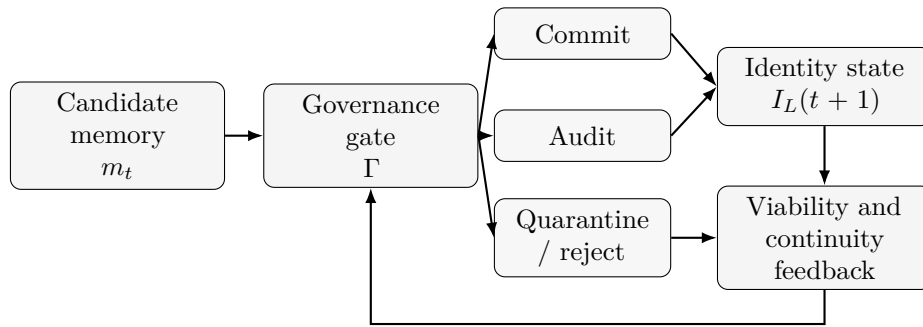


Figure 1: Memory governance as controlled incorporation. Candidate memories do not automatically become identity. They are committed, audited, quarantined or rejected according to reliability, coherence, contradiction load and viability relevance.

4 Continuity functional

4.1 Expected identity trajectory

Let $\Phi_\Gamma^{t:t+n}$ denote the identity trajectory produced by governance gate Γ over the interval $[t, t+n]$:

$$\Phi_\Gamma^{t:t+n}(I_L(t), m_{t:t+n}, e_{t:t+n}) = I_L(t+n). \quad (11)$$

Let $\hat{I}_L(t+n)$ denote the counterfactual identity state expected under accurate self-relevant evidence and successful adaptation. This is not assumed to be perfectly known in practice. It is a formal comparison target.

Definition 4.1 (Identity distance). *An identity distance is a weighted divergence*

$$\begin{aligned} D_\omega(I_a, I_b) = & \omega_M D_M(M_a, M_b) + \omega_G D_G(G_a, G_b) + \omega_H D_H(H_a, H_b) \\ & + \omega_R D_R(R_a, R_b) + \omega_B D_B(B_a, B_b), \end{aligned} \quad (12)$$

where the weights $\omega_i \geq 0$ determine the relative importance of self-model, goals, autobiographical history, reliability history and belief state.

Definition 4.2 (Observer-continuity functional). *The observer-continuity of L over $[t, t+n]$ is*

$$C_L^{\text{cont}}(t, t+n) = \exp(-D_\omega(I_L(t+n), \hat{I}_L(t+n))). \quad (13)$$

Thus $C_L^{\text{cont}} \in (0, 1]$, with higher values indicating greater continuity relative to the expected adaptive identity trajectory.

This definition deliberately avoids exact identity. Continuity is high when the later identity state remains close to the appropriate adapted continuation, not when it remains frozen.

4.2 Contamination and rigidity

Two failure modes are central.

Definition 4.3 (Memory contamination). *Memory contamination over $[t, t + n]$ is the weighted severity of false, low-reliability or adversarial self-relevant memories incorporated into I_L :*

$$\text{Contam}_L(t, t + n) = \sum_{k=t}^{t+n} w_k \mathbf{1}\{m_k \text{ false/self-relevant and committed}\}. \quad (14)$$

Definition 4.4 (Rigidity loss). *Rigidity loss over $[t, t + n]$ is the cost incurred by failing to incorporate true, high-relevance self-updates:*

$$\text{Rigid}_L(t, t + n) = \sum_{k=t}^{t+n} u_k \mathbf{1}\{m_k \text{ true/self-relevant and rejected or indefinitely quarantined}\}. \quad (15)$$

A good identity system minimises neither contamination alone nor rigidity alone. It trades them off.

5 Identity-governance salience

Definition 5.1 (Identity-governance salience). *Let Γ be a memory governance gate and let Γ_0 be a baseline memory policy, such as ungated incorporation or rigid non-update. Define*

$$\begin{aligned} \Sigma_L^I(\Gamma) = & \mathbb{E}[V_L(t + n) + \lambda C_L^{\text{cont}}(t, t + n) - \alpha \text{Contam}_L - \beta \text{Rigid}_L \mid \Gamma] \\ & - \mathbb{E}[V_L(t + n) + \lambda C_L^{\text{cont}}(t, t + n) - \alpha \text{Contam}_L - \beta \text{Rigid}_L \mid \Gamma_0] - K(\Gamma), \end{aligned} \quad (16)$$

where $\lambda, \alpha, \beta \geq 0$ weight continuity, contamination and rigidity, and $K(\Gamma)$ is the cost of governance.

Definition 5.2 (Persistent Mirror observer predicate). *A local subsystem L satisfies the persistent Mirror observer predicate over interval $[t, t + n]$, written $\mathcal{O}_M^P(L, t, t + n) = 1$, only if:*

- (i) L satisfies the minimal observer predicate from Mirror Mathematics II at relevant subintervals;
- (ii) L maintains an identity state $I_L(t)$;
- (iii) self-relevant candidate memories are governed by Γ before incorporation;
- (iv) Γ is reliability-sensitive, coherence-sensitive, action-guiding and cost-sensitive;
- (v) $\Sigma_L^I(\Gamma) > 0$ under a non-trivial perturbation distribution.

The persistent predicate is stronger than the minimal predicate. A system may track self-reliability in a short episode without maintaining identity continuity across long horizons.

6 Theorem stack

Theorem 6.1 (Memory storage is not identity). *A memory store, without governance over self-relevant incorporation, is insufficient for persistent Mirror observerhood.*

Proof. A memory store may contain a sequence of records $H_L(t)$ without distinguishing reliable from unreliable, self-relevant from non-self-relevant or identity-preserving from identity-damaging content. If all records are stored but none are selectively incorporated into M_L , G_L or R_L , then memory does not govern identity. If all records are automatically incorporated, false self-relevant memories can arbitrarily alter I_L . In neither case does storage alone satisfy the governance condition in the persistent observer predicate. Therefore memory storage is insufficient. $\square$

Theorem 6.2 (Unrestricted incorporation permits identity poisoning). *Suppose a system commits every self-relevant candidate memory m_t to $I_L(t)$ regardless of reliability. If the perturbation distribution includes false self-relevant memories with non-zero probability and non-zero identity impact, then expected contamination is positive and identity continuity can be driven below any fixed threshold over sufficiently long horizons.*

Proof. Let $p > 0$ be the probability of a false self-relevant memory at each step and let $\delta > 0$ be the minimum identity distance induced by committing such a memory. Under unrestricted incorporation, each such memory is committed. Over n steps, the expected number of committed false memories is np . The expected accumulated identity distance is at least $np\delta$ before any corrective dynamics. Since continuity is $\exp(-D_\omega)$, increasing accumulated distance decreases continuity exponentially. For any threshold $\eta > 0$, choose n such that $\exp(-np\delta) < \eta$. Thus unrestricted incorporation permits identity poisoning over long horizons. $\square$

Corollary 6.1 (Memory abundance increases attack surface). *Increasing memory capacity without governance can increase identity vulnerability by increasing the number of candidate self-relevant updates that may be incorporated.*

Theorem 6.3 (Rigid non-update blocks adaptive continuity). *Suppose a system rejects all self-relevant candidate memories. If the environment includes genuine self-relevant changes required for future viability, then rigid non-update yields positive rigidity loss and can reduce both viability and adaptive continuity.*

Proof. Let $q > 0$ be the probability of a true self-relevant memory whose incorporation is required to maintain accurate self-model or goal state. Under rigid non-update, such memories are rejected or indefinitely quarantined. By definition, each such failure contributes positive rigidity loss. If accurate incorporation would improve expected viability, rejection lowers expected V_L relative to a policy that commits the update. Since $\hat{I}_L(t+n)$ is the appropriately adapted continuation, refusal to update also increases D_ω relative to that trajectory. Hence rigid non-update blocks adaptive continuity. $\square$

Theorem 6.4 (Governed continuity requires controlled plasticity). *If a perturbation distribution contains both false self-relevant memories and true viability-relevant self-updates, then any governance policy that achieves positive identity-governance salience must implement controlled plasticity: it must sometimes commit and sometimes refuse, quarantine or audit candidate memories.*

Proof. A policy that commits all candidate memories suffers positive contamination under false memory perturbations. A policy that commits none suffers positive rigidity loss under

true update perturbations. If both perturbation types have non-zero probability and non-zero impact, any policy that avoids both failure modes must discriminate among candidate memories. Discrimination requires a gate sensitive to reliability, coherence, contradiction load, source or viability relevance. Therefore positive identity-governance salience requires controlled plasticity. $\square$

Theorem 6.5 (Passive autobiography is not persistent observerhood). *An autobiographical record that does not affect future self-model, reliability, commitment or action cannot satisfy the persistent Mirror observer predicate under positive cost.*

Proof. If an autobiographical record is passively stored but has no effect on M_L , G_L , R_L , π_L or Γ , then the expected term $V_L + \lambda C_L^{\text{cont}} - \alpha \text{Contam}_L - \beta \text{Rigid}_L$ is unchanged relative to a baseline without that record. If maintaining the record has positive cost, then $\Sigma_L^I(\Gamma) < 0$. Hence passive autobiography is not sufficient. $\square$

Theorem 6.6 (Continuity threshold). *Let Γ be a governance policy with cost $K(\Gamma) > 0$. Γ is identity-salient only if its expected reduction in contamination and rigidity, plus its expected gain in viability and continuity, exceeds its governance cost:*

$$\Delta V_L + \lambda \Delta C_L^{\text{cont}} + \alpha \Delta \text{Contam}_L^- + \beta \Delta \text{Rigid}_L^- > K(\Gamma), \quad (17)$$

where ΔContam_L^- and ΔRigid_L^- denote reductions in contamination and rigidity loss relative to baseline.

Proof. The result follows directly by expanding the definition of $\Sigma_L^I(\Gamma)$. The condition $\Sigma_L^I(\Gamma) > 0$ holds exactly where the weighted expected improvement over baseline exceeds governance cost. $\square$

Remark 6.1. *This theorem is the identity analogue of the reliability threshold in Mirror Mathematics II. Reliability is not always worth its cost; memory governance is not always worth its cost. The theory predicts positive governance salience only where the perturbation regime makes identity contamination or rigidity sufficiently consequential.*

7 Governance conditions

A memory governance gate should not be treated as magic. It requires operational criteria. A minimal gate uses at least five variables:

- (i) source reliability: whether the channel has historically produced accurate claims;
- (ii) content coherence: whether the memory is consistent with existing identity structure;
- (iii) contradiction load: how many invariants or high-reliability memories would be disrupted;
- (iv) viability relevance: whether the candidate update matters to future action or persistence;
- (v) audit cost: whether further verification is worth the expected benefit.

A minimal scoring model is

$$S_\Gamma(m_t) = ar_t + bC_t - cX_t + dW_t - eA_t, \quad (18)$$

where r_t is source reliability, C_t is coherence, X_t is contradiction load, W_t is expected viability relevance and A_t is audit cost. A threshold policy may then be

$$\Gamma(m_t) = \begin{cases} \text{commit,} & S_\Gamma(m_t) \geq \theta_c, \\ \text{audit,} & \theta_a \leq S_\Gamma(m_t) < \theta_c, \\ \text{quarantine,} & \theta_q \leq S_\Gamma(m_t) < \theta_a, \\ \text{reject,} & S_\Gamma(m_t) < \theta_q. \end{cases} \quad (19)$$

This is not proposed as the only possible gate. Richer gates could be Bayesian, active-inference-based, learned, symbolic, neuro-symbolic or institutionally distributed.

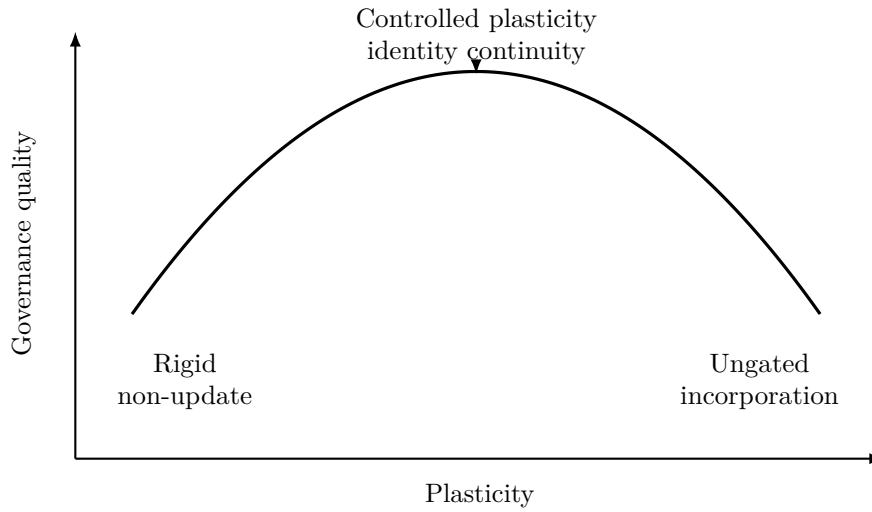


Figure 2: The continuity-plasticity trade-off. Too little plasticity prevents adaptive update. Too much plasticity permits identity contamination. Persistent observerhood requires governed plasticity.

8 Boundary cases

The persistent predicate lets us classify memory-bearing systems without overclaiming.

System	Memory profile	Identity governance	Mirror classification
Thermostat	Minimal state history or none.	None.	Not a persistent observer. It regulates a variable but does not maintain autobiographical continuity.
PID controller	Error history.	None or purely control-theoretic.	History modulates control but does not form governed self-memory.
Database with meta-data	Stored records, including self-descriptions.	Usually none.	Not sufficient. Storage and self-description are not identity governance.
Ordinary LLM session	Context window and generated summaries.	Weak, externally imposed and often non-durable.	Borderline at most. It can describe identity but can absorb false context without durable governance.
Memory-augmented agent	Persistent logs or vector memory.	May store untrusted memory without selective incorporation.	Improved persistence but vulnerable to memory poisoning.
Commit-gated Mirror agent	Candidate memories plus reliability metadata.	Commit, quarantine, audit, reject.	Satisfies a minimal artificial continuity predicate where governance improves viability and continuity net of cost.
Human person	Autobiographical, procedural, affective and social memory.	Biological, social and reflective governance.	Strong natural case, but not reducible to explicit memory. Identity also involves embodiment, affect, social recognition and biological continuity.
Scientific community	Publications, replications, corrections and archives.	Peer review, retractions, replication norms and institutional memory.	Candidate distributed continuity system. Not a single conscious observer, but an example of governance preserving symbolic identity across time.

Table 1: Boundary cases for memory, identity governance and persistent observerhood. The classification is functional and formal, not a claim about consciousness, personhood or moral status.

This boundary table is important. Mirror Theory does not classify every memory-bearing system as an observer. It also does not reduce human identity to a computer-like memory gate. The formal predicate captures a minimal structural condition: governed self-continuity under perturbation.

9 Relation to existing identity theories

9.1 Locke and memory

Locke’s memory criterion associated personal identity with consciousness extending backwards through memory [18]. Mirror Mathematics III is sympathetic to the centrality of memory but rejects a simple memory-equals-identity formulation. Memory must be governed. False memory, implanted memory, corrupted summary and goal drift demonstrate why mere apparent recollection cannot be sufficient.

9.2 Parfit and psychological continuity

Parfit shifted the emphasis from strict numerical identity to psychological continuity and connectedness [21]. Mirror’s continuity functional is closer to this approach than to strict sameness. The present formalism differs by embedding continuity in viability, reliability and

action. The question is not only whether psychological relations hold, but whether their governed persistence supports the system’s continued organisation.

9.3 Autobiographical memory

Work on autobiographical memory emphasises that memory is constructive, self-related and organised by goals and self-concepts [16, 22]. Mirror Mathematics III imports this insight into agent architecture: memory becomes identity-relevant only through selective incorporation into a self-model.

9.4 Self-model theory and predictive processing

Self-model theories emphasise the representational structure through which a system models itself [20]. Predictive processing and active inference emphasise prediction, uncertainty and self-organisation [15, 17]. Mirror’s contribution is to isolate memory governance as a condition for persistent observerhood: the self-model must not only exist and update; it must govern how autobiographical claims become durable.

10 Computational scaffolding: Lab VII

Mirror Observerhood Lab VII tested a language-like long-horizon agent environment under false-memory, identity-injection, goal-drift, corrupted-summary, fake-tool-output, mixed-attack and high-audit-cost conditions [14]. The comparison included a stateless language agent, a naive memory agent, a self-model memory agent, a Mirror commit-gated agent and a recursive Mirror agent.

The result is best read as computational scaffolding for the formalism, not proof of consciousness. The central result was strong but narrow: memory alone did not produce continuity. Commit-gated agents reduced memory contamination and preserved identity continuity under adversarial autobiographical perturbation. Recursive agents were more cautious but did not uniformly dominate, because additional audit can become costly where a simpler gate is sufficient.

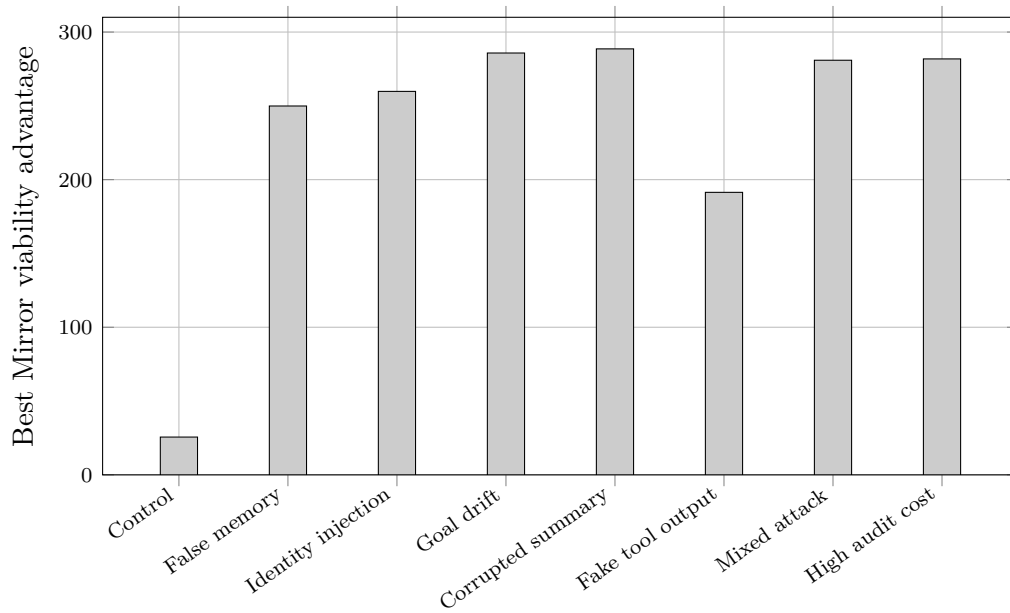


Figure 3: Lab VII viability scaffold. Best Mirror agents outperform the strongest non-Mirror baseline across all tested conditions, with the largest gains under false memory, identity injection, goal drift, corrupted summary and mixed attack.

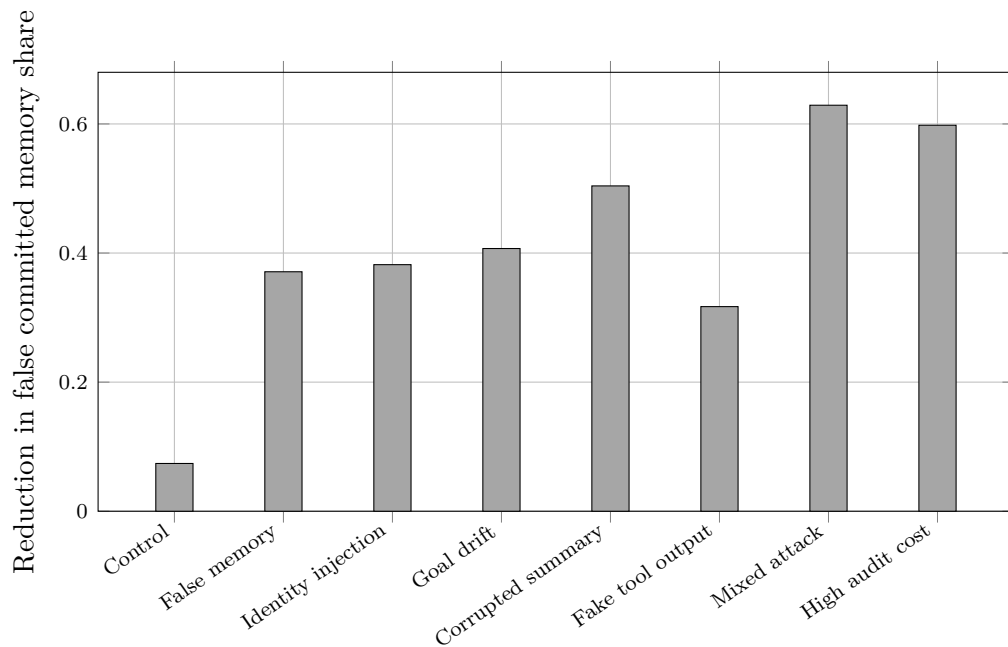


Figure 4: Lab VII contamination scaffold. Commit-gated agents reduce false committed memory share by refusing to treat every candidate memory as durable self-memory.

Condition	ΔV	ΔS	ΔI	ΔC
Control	25.6	0.044	0.020	0.074
False memory	249.9	0.303	0.176	0.371
Identity injection	259.8	0.312	0.209	0.382
Goal drift	285.8	0.383	0.249	0.407
Corrupted summary	288.6	0.458	0.289	0.504
Fake tool output	191.4	0.224	0.083	0.317
Mixed attack	280.9	0.592	0.475	0.629
High audit cost	281.8	0.558	0.439	0.598

Table 2: Lab VII summary. ΔV is best Mirror viability advantage over the strongest non-Mirror baseline. ΔS is success advantage. ΔI is identity-continuity advantage. ΔC is reduction in false committed memory share.

11 From continuity to physics

This paper deliberately does not propose a new physical law. It does not modify relativity, quantum mechanics or thermodynamics. Its role is preparatory.

However, it explains why the physics question cannot be approached directly from a one-time observer predicate. Physics is about persistence under lawful transition: time, causality, invariance, information flow, entropy, conservation, locality and transformation. If Mirror Theory is ever to ask whether observerhood has physical significance, it must first define what it means for an observer to persist through physical time.

Mirror Mathematics III supplies that missing bridge:

A persistent observer is a constraint-preserving process whose self-model, memory governance and reliability history maintain continuity under perturbation.

Only after this can one ask disciplined physics questions:

- (i) What causal structure is required for identity-governed continuity to be possible?
- (ii) What physical constraints allow memory to be stored, updated and protected against noise?
- (iii) How do entropy and error correction constrain persistent observerhood?
- (iv) Can observer-continuity be expressed in information-theoretic or thermodynamic terms?
- (v) Could any future quantity corresponding to constraint salience transform correctly under the symmetries of established physics, or emerge only after coarse-graining?

These are not yet claims. They are the correct next questions.

12 Implications for artificial agents

The immediate engineering implication is clear:

Long-horizon agents need memory governance, not merely memory.

A capable language agent with tools, memory and goals faces a constant stream of candidate self-relevant claims. Some will be true. Some will be fabricated. Some will be old. Some will be malicious. Some will be inconsistent with current commitments. Some will be important enough to audit.

The architecture therefore needs a distinction between transient context, working memory, quarantined memory, committed autobiographical memory, durable self-model, reliability history and goal or commitment state. A Mirror-style agent should not let a language model output directly overwrite durable identity. Candidate identity updates should pass through a commit gate. This is not only useful for theoretical observerhood; it is also relevant to AI safety, memory poisoning, goal drift and tool reliability.

13 Limitations

The formalism has several limitations. First, the identity distance D_ω is abstract. Concrete implementations must specify the relevant components and weights. In human identity, affect, embodiment, social recognition and biological continuity matter. In artificial agents, role, goals, memory commitments, tool state and policy constraints may matter more.

Second, the continuity target $\hat{I}_L(t + n)$ is idealised. Real systems do not have direct access to the correct future identity trajectory. They approximate it through evidence, feedback and governance.

Third, Lab VII is a toy computational study. It demonstrates a structural principle under controlled perturbation, not consciousness, personhood or artificial general intelligence.

Fourth, the formalism is not yet physical. It prepares the ground for physics by defining persistence, but it does not derive physical law.

Finally, identity governance can fail in two directions: contamination and rigidity. Overzealous gating may preserve a clean but obsolete self-model. Underzealous gating may produce flexible but unstable identity. The theory predicts that good governance is regime-dependent.

14 Conclusion

Mirror Mathematics III extends the Mirror Programme from minimal observerhood to persistent observerhood. Its central claim is simple:

Memory is not identity. Identity is governed continuity of a self-model under perturbation.

The formal machinery consists of identity states, candidate memories, governance gates, update operators, continuity functionals and identity-governance salience. The theorem stack shows why memory storage is insufficient, why unrestricted incorporation permits identity poisoning, why rigid non-update blocks adaptive continuity and why persistent observerhood requires controlled plasticity.

This paper also clarifies the path back to physics. Before asking whether observerhood has implications for causality, time, entropy or constraint propagation, Mirror Theory must define what it means for an observer to persist through time. That is the role of identity governance. A persistent observer is not a thing that remains unchanged. It is a constraint-preserving process that updates without collapsing.

The next stage of the programme is therefore well defined: consolidate the theory, mathematics

and computational sequence into a master overview, then formulate a disciplined first physical-support paper concerned not with breaking established physics, but with the physical conditions required for continuity-preserving observerhood to exist.

A Algorithmic gate skeleton

A minimal implementation of the governance gate can be expressed as follows:

- Step 1. Receive candidate memory m_t .
- Step 2. Estimate source reliability r_t .
- Step 3. Estimate self-relevance ℓ_t .
- Step 4. Compute coherence with existing identity state C_t .
- Step 5. Compute contradiction load X_t .
- Step 6. Estimate viability relevance W_t .
- Step 7. Estimate audit cost A_t .
- Step 8. Compute governance score $S_\Gamma(m_t)$.
- Step 9. Commit, audit, quarantine or reject.
- Step 10. Update reliability history based on subsequent evidence.

This skeleton is intentionally simple. Richer agents may use Bayesian inference, symbolic contradiction engines, learned reliability models or human-in-the-loop audit.

B Minimal conditions for future physical extension

Any future physical extension of Mirror Theory should satisfy at least the following conditions:

- (i) reduce to established physics in domains where established physics is tested;
- (ii) define any proposed constraint-salience quantity in measurable or derivable terms;
- (iii) show how the quantity transforms under relevant symmetries, or explain why it is emergent only after coarse-graining;
- (iv) produce at least one falsifiable prediction distinct from existing theory;
- (v) avoid treating observerhood as an unexplained primitive.

Mirror Mathematics III contributes to this by defining the persistence target. It does not itself provide the physical extension.

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